

Homework Set 2, CPSC 6430/4430

LastName, FirstName

Due 03/13/2026, 11:59PM EST

GD for Ridge Regression

For Ridge Regression:

$$\min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{y}\|_2^2 + \gamma \|\mathbf{x}\|_2^2, \quad (1)$$

1. What is the derivative with respect to $\mathbf{x}$?
2. What is the closed solution of ridge regression?
3. Assume $\mathbf{A} \in \mathbb{R}^{100 \times 50}$, and stepsize $\lambda = 0.5/(\gamma + \sigma_1(\mathbf{AA}^T))$, where $\sigma_1(\mathbf{Z})$ denotes the largest singular value of $\mathbf{Z}$. Please change λ from $\{0.01, 0.1, 1, 10, 100\}$ and plot the objective changes with update via gradient descent method (100 iterations), what do you find?

Optimization for Lasso

We now turn to discuss Lasso model which is not mentioned in class:

$$\min_{\mathbf{x}} \frac{1}{2} \|\mathbf{Ax} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{x}\|_1, \quad (2)$$

where we define $\|\mathbf{z}\|_2^2 = \sum_i \mathbf{z}_i^2$, $\|\mathbf{x}\|_1 = \sum_i |\mathbf{x}_i|$. Different from linear regression or ridge regression, where we can obtain the optimal solution by taking the derivative with respect to $\mathbf{x}$ and set to 0, the main drawback of Lasso model is $\|\cdot\|_1$ is not differentiable. To address the issue, we turn to a different optimization framework which is summarized as below:

Algorithm 1 An algorithm to solve Eq. (2)

```
 $\mathbf{x} \leftarrow 0$ 
 $t \leftarrow 1/\sigma_1(\mathbf{AA}^T)$ 
 $k \leftarrow 1$ 
 $K \leftarrow 100$ 
while  $k \leq K$  do
     $\mathbf{z} = \mathbf{x} - t\mathbf{A}^T(\mathbf{Ax} - \mathbf{y})$ 
    ***for each element in  $\mathbf{z}$  and  $\mathbf{x}$ ***
     $\mathbf{x}_i = \text{sign}(\mathbf{z}_i) * \max\{|\mathbf{z}_i| - t\lambda, 0\}$ 
end while
```

1. Please implement the algorithm and plot the objective changes with update (for $\lambda = 1$ and $\lambda = 2$, respectively) where x-axis is the iteration and y-axis is the objective in Eq. (2).
2. Please compare the output from Algorithm 1 with Lasso in sklearn and check whether they are close. You can randomly generate $\mathbf{A} \in \mathbb{R}^{100 \times 50}$, $\mathbf{y} \in \mathbb{R}^{100}$.